Charlotte Morissette

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EDUCATION

PhD, Robotics - University of Toronto **Sept 2026 - Present**
Supervisor: Jonathan Kelly *Toronto, Canada*

MSc Thesis, Computer Science - McGill University **Sept 2023 - Feb 2026**
Thesis: Tactile Modality Fusion of Vision-Language-Action Models *Montréal, Canada*
Supervisor: Gregory Dudek - *CGPA:* 4.00/4.00

BSc, Honours Computer Science & Biology - McGill University **Sept 2019 - May 2023**
Minor: Statistics - *CGPA:* 3.80/4.00 *Montréal, Canada*

WORK EXPERIENCE

Samsung AI Center *Montréal, Canada*
Research Intern: ***May 2022 - Aug. 2023***

- *Human-Robot Handover:* Investigated risk-dependent human handover behaviour using high-resolution tactile sensing, analyzing timing and tactile negotiation during hazardous versus safe object transfers to inform safer robot policies. **NeurIPS.**
- *Visual-Tactile 2.5D Shape Inference:* Developed a visual-tactile method for contact-shape estimation based on changes in tactile marker separation, enabling shape inference across sensor embodiments without retraining. Built tactile sensors, collected validation data, and contributed to method development and writing. **IEEE RA-L**.
- *Hypernetworks for Zero-Shot RL:* Developed a hypernetwork-based approach for zero-shot transfer across variations in task rewards and physical dynamics. Designed and implemented continuous-control environments and contributed to theory and writing. **AAAI.**

McGill University *Montréal, Canada*
Teaching Assistant ***Sept 2024 - May 2025***

- TA for COMP462 & COMP561, Computational Biology Methods and COMP421, Database System
- Led tutorials, prepared course materials, and guided students on assignments and course concepts

Research Assistant ***Sept 2020 - Aug 2021***

- Research position in Prof. Joseph Vybihal's lab at McGill University
- Investigated neural network classification in low-data regimes and explored deep learning approaches for image inpainting.

PUBLICATIONS

2026 **Morissette, C.**, Abyaneh, A., Chang, W-D., Houssaini, A., Meger, D., Lin, H., Tremblay, J. and Dudek, G."Tactile Modality Fusion for Vision-Language-Action Models" *In European Conference on Computer Vision (ECCV) 2026.*

- Developed **TacFiLM**, a novel lightweight method for integrating tactile sensing into VLA models. Led the full pipeline from architecture design and training to real-robot evaluation, achieving improved manipulation performance. **ECCV 2026.**

2026 Abyaneh, A., **Morissette, C.**, Danesh, M., Houssaini, A., Meger, D., Dudek, G. and Lin, H. 2025. "Contractive diffusion policies: Robust action diffusion via contractive sampling with differential equations." In International Conference on Learning Representations (ICLR) 2026.

- Proposed a contractive sampling method to improve robustness and reduce action variance in diffusion-based control policies. Implemented core components and led real-world experiments on a Franka arm. **ICLR 2026.**

2025 Wen, S., Meriaux, E., Guzmán, MS., **Morissette, C.**, Si, C., Baghi, B. and Dudek, G. "Scalable Aerial GNSS Localization for Marine Robots". In *IEEE International Conference on Robotics and Automation (ICRA), Robots in the Wild Workshop 2025.*

- Introduced a method to localize marine robots via GNSS-equipped aerial drones, avoiding the signal loss and cost of onboard aquatic GNSS. Led field robot data collection and experiments. **ICRA 2025.**

2024 Jilani, A., Hogan, F.R., **Morissette, C.**, Dudek, G., Jenkin, M. and Siddiqi, K. "Visual-Tactile Inference of 2.5D Object Shape from Marker Texture". In *IEEE Robotics and Automation Letters 2024, presented at ICRA 2025.*

- Description in work experience

2023 Rezaei-Shoshtari, S., **Morissette, C.**, Hogan, F.R., Dudek, G. and Meger, D. "Hypernetworks for Zero-shot Transfer in Reinforcement Learning". In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 37), 2023.

- Description in work experience

2022 **Morissette, C.**, Baghi, B.H., Hogan, F.R. and Dudek, G. "A Study of Human-Robot Handover through Human-Human Object Transfer". In *Advances in Neural Information Processing Systems (NeurIPS), Human in the Loop Learning (HiLL) Workshop 2022.*

- Description in work experience

PATENTS

2024 Jilani, A., Hogan, F.R., **Morissette, C.**, Dudek, G., Jenkin, M. and Siddiqi, K. "Optical tactile sensor and method for estimating shape from touch". United States Patent US 18,378,447. United States Patent and Trademark Office. 19 Sept. 2024

2024 Rezaei-Shoshtari, S., **Morissette, C.**, Hogan, F.R., Dudek, G. and Meger, D. "Hypernetworks for Zero-shot Transfer in Reinforcement Learning". United States Patent US 18,385,696. United States Patent and Trademark Office. 20 Jun. 2024

TECHNICAL SKILLS

Programming: Python, Java, C++, C, SQL, Javascript, Bash
ML Frameworks: PyTorch, Jax, Hugging Face, Numpy, Pandas, Scikit-learn, Keras, Tensorflow
Robotics: ROS, Gazebo, Mujoco, Bullet, Isaac Sim/Lab, MoveIt
Tools: Docker, Git, Google Cloud, Jupyter, CAD/fabrication
Languages: French, English

SELECT HONOURS AND AWARDS

2026-Present Fonds de Recherche du Quebec (FRQ-NT) - Doctoral Research Scholarship.

2024-2025 Fonds de Recherche du Quebec (FRQ-NT) - Master's Research Scholarship.

2023 AAAI-23 Student Scholarship.

2020-2021 Faculty Of Science Scholarships, Top 5% of the Faculty.

EXTRACURRICULAR ACTIVITIES

Volunteer, PSSAR Network: Supporting Students and Scholars at Risk **May 2024 - Present**
- Match students with professors and graduate programs in Canada. *Montréal, Canada*
- Assist with graduate school applications.

Volunteer, Women in AI & Robotics **Jun 2022 - Jan 2024**
- Core member; helped found the WAIR Youth Group and organized robotics hackathons. *Montréal, Canada*

Session Chair, AAAI conference on artificial intelligence **Feb 2023**
- Chaired two sessions on *ML: Deep Neural Architectures* during 2023 conference *Washington DC, USA*